

Avi Shukla

6393906196 | avishuklawork@gmail.com | linkedin.com/in/avi-shukla | github.com/xOAviOx

EDUCATION

BML Munjal University

B.Tech in Computer Science Engineering

Gurgaon, Haryana

Aug. 2025 – May 2029

Delhi Public School

Senior Secondary (PCM + CS)

Lucknow, Uttar Pradesh

Aug. 2022 – May 2024

EXPERIENCE

Back End Developer Intern

Mar 2026 – May 2026

Stealth Startup

Remote

- Built REST API endpoints in Node.js (Express) and Python for an early-stage product, working directly with the founding team to take features from idea to production
- Designed the PostgreSQL schema with indexing and query optimization, and added Redis caching for sessions and frequently-hit read paths to keep response times low as usage grew
- Set up the server-side essentials ahead of the beta — request validation, structured error handling, and logging

Full Stack Developer Intern

Aug 2025 – Oct 2025

Dendrite.ai

Remote

- Built full-stack features for an ML + BI analytics platform used by data scientists and quant researchers — React/TypeScript on the front end, REST APIs behind it
- Reworked the data pre-processing and transformation pipelines behind the research workbench, cutting prep time on common workflows by around 30%
- Worked on backend services for model training, experiment tracking, and deployment, leaning on pagination and caching to keep large datasets responsive

Full Stack Developer Intern

Mar 2025 – May 2025

Messold Technologies

New Delhi

- Built responsive e-commerce storefronts for D2C brands with React, Next.js, and TypeScript, pushing Lighthouse performance scores from the low 60s into the 90s
- Built campaign analytics dashboards (Node.js, PostgreSQL, REST APIs) that surfaced ad performance and store metrics across several client accounts
- Integrated third-party APIs (Shopify, Meta Ads, Google Ads, payment gateways) and webhook pipelines to automate order tracking and attribution

PROJECTS

Maestro | [GitHub](#) | [Demo](#) | *Electron, TypeScript, React, Node.js, Zustand, SQLite, Git*

June 2026 – Present

- Built a cross-platform Electron desktop app (Windows/macOS) that runs multiple CLI coding agents in parallel, each isolated in its own Git worktree, with a single UI to review and merge everything they produce
- Designed a security-hardened multi-process setup — sandboxed renderer with context isolation, a typed IPC bridge, and Zod-validated IPC payloads — keeping all filesystem, Git, and child-process access in the main process
- Wrote the orchestration engine in strict TypeScript (DAG-based task scheduler, workspace supervisor, concurrent agent lifecycles), with a Monaco diff viewer, merge/PR workflows, and embedded node-pty + xterm.js terminals for live agent output

Gantry | [GitHub](#) | [Demo](#) | *Go, Next.js, TypeScript, PostgreSQL, Docker, Caddy*

July 2026 – Present

- Built a single-node Platform-as-a-Service in Go and Next.js that turns a `git push` into a live app on its own subdomain — handling Docker image builds, container health checks, and Caddy reverse-proxy routing, with build logs streamed live over Server-Sent Events (SSE)
- Implemented zero-downtime blue/green deployments with one-click rollback on a PostgreSQL-backed job queue (SKIP LOCKED claims, worker heartbeats, crash recovery)
- Added a self-healing reconciliation loop that restores killed containers and broken routes within seconds, AES-256-GCM encrypted secrets management, and HMAC-verified GitHub webhooks for push-to-deploy CI/CD

AICut | [GitHub](#) | *Python, FastAPI, React, TypeScript, Whisper, Ollama, FFmpeg*

July 2026 – Present

- Built a local-first, AI-controlled desktop video editor (FastAPI + React/TypeScript) that treats the transcript as the timeline — you cut footage by deleting words or typing natural-language commands, with instant skip-based preview and no re-rendering
- Designed a hallucination-safe LLM pipeline: a local model (Ollama) emits typed, schema-validated JSON edit plans that reference transcript word IDs and compile deterministically into cut ranges, so the model never invents timestamps
- Built the media pipeline end to end: GPU-accelerated speech-to-text with faster-whisper (word-level timestamps, English/Hindi), and FFmpeg export with burned-in captions, 9:16 reframing, NVENC encoding, and SSE-streamed progress

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, C/C++, Java, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js, Express, FastAPI, Electron, Tailwind CSS

AI/ML: OpenAI & Gemini APIs, Ollama, LangChain, CrewAI, AutoGen, Whisper (faster-whisper), PyTorch, OpenCV, FFmpeg, RAG

Databases & Infra: PostgreSQL, MongoDB, Redis, SQLite, Supabase, Docker, Caddy, Nginx, Ubuntu (VPS)

Concepts & DevOps: REST APIs, Webhooks, SSE, CI/CD, Blue/Green Deploys, Job Queues, AES-256-GCM, OAuth, HMAC, Git, Vercel, Render, n8n, Postman